Field of computer Science & Engineering

Data Mining

GALALA UNIVERSITY
Powered by Arizona State University®

Project Title *(English)*	**Crime Pattern and Behavior Analysis**	
Team and ID	**Hala Ahmed** 222102480 **Malak Amgad** 221100451 **Mena Osman** 221100521 **Menna Salem** 221101277 **Sama AbdelTawab** 221100879 **Arwa Ahmed** 221100209	
Field	Computer Science and Engineering	
Program	Artificial Intelligence & Computer Engineering	
Instructor	**Dr. Mohamed Abd Elaziz**	
This part for instructor: Notice		**Degree**
Date of Submission		

Content

A.Abstract .....

B. Introduction.....

C. Objectives.....

G. System Analysis and Desgin .....

H. Implementation and Outputs.....

I. Conclusion.....

Field of computer Science & Engineering

Data Mining

GALALA UNIVERSITY
Powered by Arizona State University®

Abstract project:

Crime has been on the rise in the past few years. Several research papers try to correlate and understand the reason for this increase; however, it still remains unclear which factor affects the crime rate. Therefore, this study aims to understand the crime patters in Los Angeles and find the most common patterns in the last few years by utilizing different techniques such as clustering, descriptive analysis, time-series analyses, and predictive modeling. We utilize more than a million-crime record and relate it to the socioeconomic and demographic features for the area it occurred in. Results show that crime peak on weekdays around 12 at the afternoon and it occur commonly in areas like 77th street. K-Means clustering show the community clusters differentiated by crime intensity and socioeconomic status. Additionally, HistGradientBoosting achieved 82% in hotpot forcasting and lighGBM reached 75% in severity prediction while in regression XGboost reached a MSE of 0.22 in predicting the crime code.

Field of computer Science & Engineering

Data Mining

GALALA UNIVERSITY
Powered by Arizona State University®

Introduction and related work:

As years pass by, there has been an increase in the crime rates around the world especaily in highly diverse and populated countries like Unites States of America. The crimes happen due to various reasons ranging from hate crimes to accedental crimes. However, most of the crimes have underlying patterns that can be realted to socioeconomic and demographic factors that can be discovered by various technqiues such as data mining. The techniques help in gaining insights related to crime patterns which helps in identifying and lowering the crime rate.

Many states and areas in the United States of America suffer from high and unregulated crime rates especially Los Angeles (LA). Los Angeles is a city in the California state and is considered the second most populous in the United states reaching around 4 million residents. It is highly diverse and include different cultures, backgrounds, and economic classes. Due to these different factors, LA is known to experience high crime rates surpassing by 24.7% higher than the national average. Therefore, it has become quite essential to determine the patterns that crimes and to identfy the high risk zones to ensure proper prevention and understand how to lower the rate.

Recent literature highlights the growing effectiveness of data mining techniques in crime research showcasing both methodological diversity and quantifiable performance improvements. Deepika K.K. and Smitha Vinod (2018) integrated K-means clustering with Random Forest and neural network classifiers, achieving 92.4% accuracy in classifying major offence categories, while also generating visually distinct clusters using Google Maps-based analytics. In another study, Yerupude and Gudur (2017) applied decision trees to explain 68% of the variance in district crime rates and achieved 81% accuracy using a Naïve Bayes classifier for headline crime prediction. A systematic review by Kaur and Williamjeet Singh (2017) confirmed the effectiveness of frequent pattern mining in retrieving 93% of known crime sequences, while their outlier detection algorithm maintained a low false alarm rate of just 0.14..

In the following project, several datasets were collected and merged to ensure that several factors is taken into considereaion. There is a primary dataset that was mainly used and there were 7 other datasets that was taken from the same source. The datasets were all connected from government websites such as LA county and . The primary dataset inlcuded 28 features that was collected from the span of 2020 to 2023. The features describe the details of each crime comitted such as the victim age, weapon used, crime commited and etc. The data included over 1 million records of crimes commited which allowed us to apply several technqiues and ensure proper analysis and prediction. The additional datasets included features that were

describing the 74 areas in Los Angeles and it included details regarding health of residents, crimes rates, education, and etc. By utilizing the following features, we were able to understand if the following socioeconomic and demographic features affect and influence the crime ocrruances and to not just depend of the crime occurances details but also look at other factor.

In the following project, several data mining techniques will be used to analyze the crime incidents in LA and show the patterns that are associated with it. Several steps will be applied to ensure correct analysis including data merging and data preprocessing to ensure the data is properly cleaned after merging and to not allow any mistakes to inclfluence the results. Afterwards, descriptive analysis and time-series analysis is applied to identidy temporal trends and patterns related to socioeconomic and demographic patterns. Then, clustering and predictive modeling is applied to predict hotpost areas and group similar clusters. Finally, associate rule mining was done to understand patterns that show up together.

Field of computer Science & Engineering

Data Mining

GALALA UNIVERSITY
Powered by Arizona State University®

Objectives:

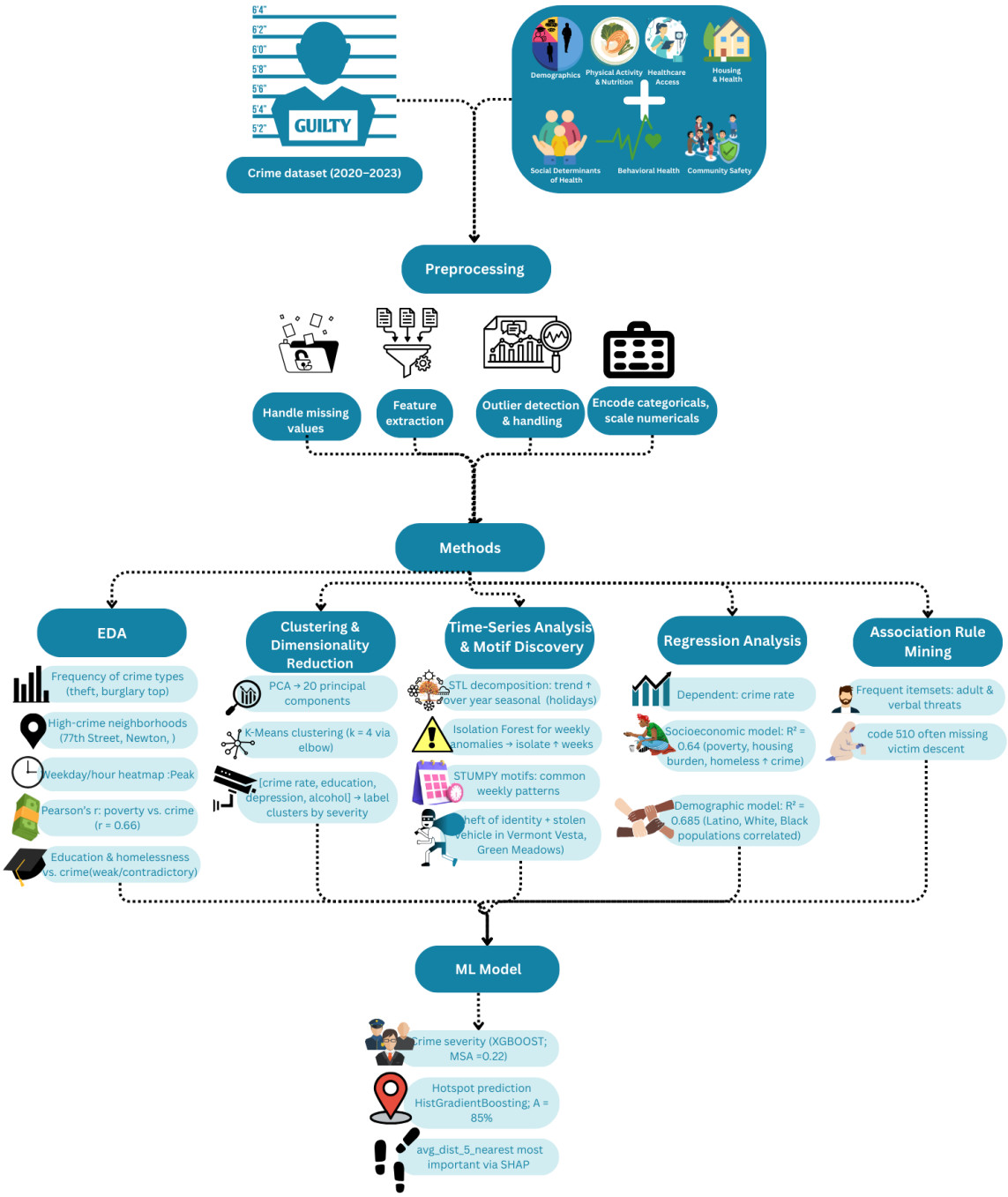
1) *Find patterns that show up with different crimes.*
2) *Determine whether an area will be a crime hotspot.*
3) *Determine the effect of socioeconomic and demographic factors on crime*
4) *Predict the severity of crime and the crime code that was committed.*
5) *Determine what patterns show up together in certain crimes.*

Field of computer Science & Engineering

Data Mining

GALALA UNIVERSITY
Powered by Arizona State University®

System Analysis and Design:



The project goes through several phases to achieve the objectives that were mentioned above. It follows a specific workflow which can be seen below:

1) **Data Merging:** The secondary datasets were filtered based on what features we will need. Then, the datasets were merged and the instances that were not describing the areas in Los Angeles were removed, and we defined the longitude and latitude of the areas and a feature was applied to find the nearest area that the crime occurred in. Then, the data was merged with the primary dataset to it.
2) **Data Pre-Processing:**
   a) **Missing Values:** The categorical features were replaced with "none" or "unknown". The numerical features were either replaced with their mean value or they were dropped.
   b) **Outlier Detection & Analysis:**
   c) **Datetime transformation:** Extracted several features were such as weekday, hour, season, and shift from the time and date of occurrence.
   d) **Feature Encoding and Feature Scaling:** the categorical features where all label and min-max normalization was applied to ensure that all values are in the same range.
3) **Exploratory and Descriptive Analysis:**
   - Summaries and visualization were implemented to determine the crime frequency, location, victim profiles, temporal trends, and correlation with socioeconomic and demographic features.
4) **Dimensionality Reduction and Clustering:**
   - PCA was applied to reduce the data into 20 components and to determine which features hold most of the information
   - K-Means clustering was utilized to cluster neighborhood according to crime severity count and applied time-based clustering and victim profiling.
5) **Time-Series Analysis:**
   - Applied Isolation Forest for anomaly detection
   - STL was used to understand trends, seasonality, and residual.
   - Motif discovery was applied to detect repeating crime patterns.
6) **Predictive Modeling:**
   - **Crime Severity Classification**: LightGBM was used to predict severity.
   - **Hotpot Prediction:** HistGradientBoostingClassifier predicted high-crime zones.

- **Crime Rate Regression:** OLS regression was used to understand how socioeconomic and demographic data.

7) **Associate Rule Mining:**

- Apriori Algorithm was applied on victim age, gender, and weapon data.

Field of computer Science & Engineering

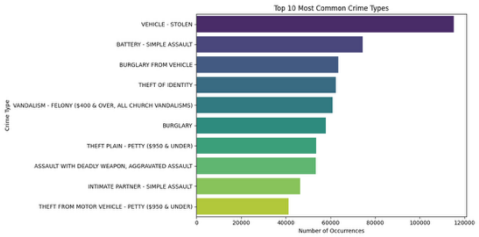
Data Mining

GALALA UNIVERSITY
Powered by Arizona State University®

Implementation and Outputs:

1) **Descriptive Analysis:** After pre-processing was applied, several visualizations were applied to understand our data and to visualize some of the patterns in the occurrences of crimes. They can be divided into the following categories

   a) **Most common crime type:** After performing several visualizations as shown in the figure below. The most common crime types include crime related to theft and burglary and are more common than crimes related to assault.

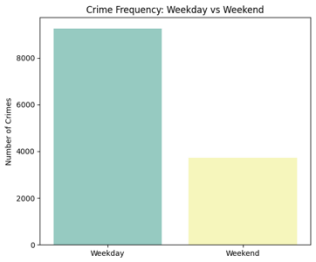


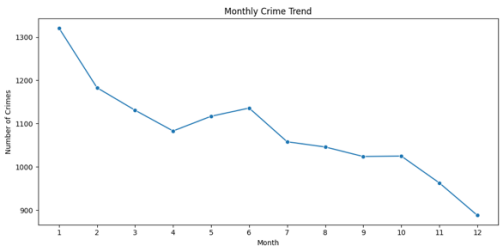
   b) **High-Crime Neighborhoods:** it was discovered that crimes most commonly happen in 77th street and then Newton and Central.

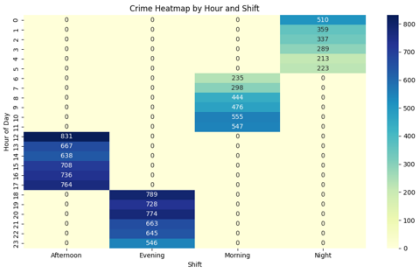
```
# Mean Crimes per Neighborhood
crime_by_area = data_cleaned.groupby('AREA NAME')['Crimes_m'].mean().sort_values(ascending=False)

plt.figure(figsize=(12, 6))
sns.barplot(x=crime_by_area.values, y=crime_by_area.index, palette="magma")
plt.title("Average Crime Rate by Neighborhood")
plt.xlabel("Average Crimes_m")
plt.ylabel("Neighborhood")
plt.tight_layout()
plt.show()
```

   c) **Temporal Trends:** several analyses were done to show the crimes relation to temporal trends where it can be inferred from the figures below that crimes happen usually in the afternoon or the evening on weekdays. Also, crimes usually happen at the beginning of the year and then fall off.

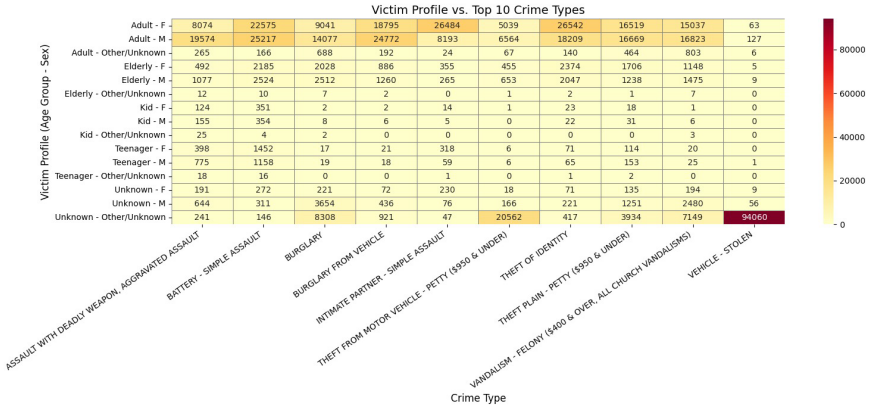




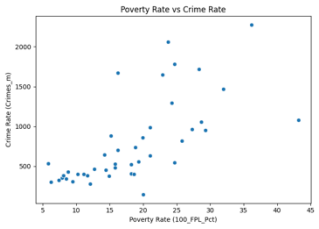


   d) **Age and Gender Analysis:** it was then visualized how the age and sex of the victim related to the crime committed as shown below. Where it was determined the most common crime was stealing the vehicle for unknown gender and sex. Additionally, for female victims the

most common crimes were simple assault, theft of identity, and battery. While male victims often face battery, vehicle burglary, and aggravated assault with weapon.



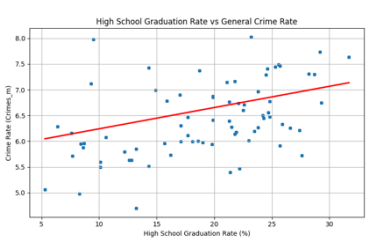
e) **Poverty vs. Crime:** it was revealed that in poorer neighborhoods that they have a higher crime rate due to the Pearson coefficient reaching 0.66.

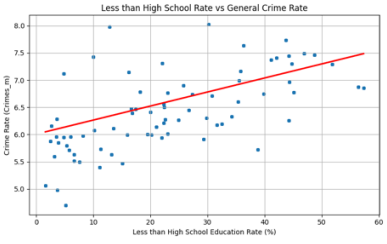


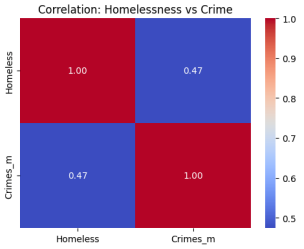

```
plt.figure(figsize=(7, 5))
sns.scatterplot(data=data_cleaned, x='100_FPL_Pct', y='Crimes_m')
plt.title("Poverty Rate vs Crime Rate")
plt.xlabel("Poverty Rate (100_FPL_Pct)")
plt.ylabel("Crime Rate (Crimes_m)")
plt.tight_layout()
plt.show()

# Pearson correlation
corr, pval = pearsonr(data_cleaned['100_FPL_Pct'], data_cleaned['Crimes_m'])
print(f"Pearson Correlation (100_FPL_Pct vs Crimes_m): {corr:.3f} (p-value: {pval:.3f})")
```

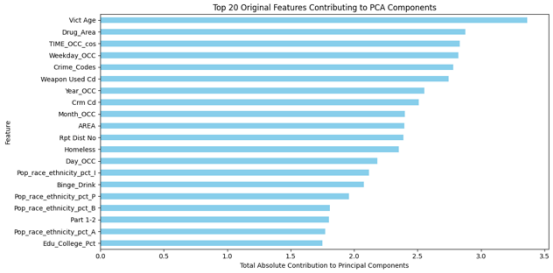
f) **Education and Homelessness vs. Crime Rate:** As shown in the figures below, it seems that both figures show that as high school graduates increase, crime rates also increases. While the others show than area with less high school rate, had more crime. This seems contradictory which shows the education on its own does not affect crime rate. Additionally, there is weak correlation between homelessness and crime rates.







2) **Dimensionality Reduction:** Dimensionality reduction was then applied to the datasets so we it would aid in understanding the structure of the datasets and determine which features hold the most important information. The technique that was used was PCA and it reduced the dimensions to 20 components. The main components is as shown below:

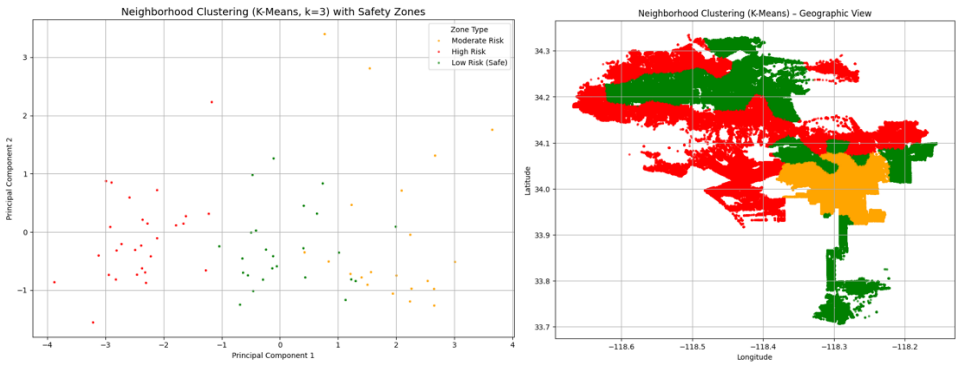



```
pca = PCA(n_components=20)
X_pca = pca.fit_transform(X_scaled)

pca_df = pd.DataFrame(X_pca, columns=[f'PC{i+1}' for i in range(20)])
loadings = pd.DataFrame(pca.components_.T,
                        index=numerical_df.columns,
                        columns=[f'PC{i+1}' for i in range(20)])
```

3) **Clustering:** Several clustering techniques were used to identify the clusters throughout the dataset. The main points we concentrated on were:

   a) **Neighborhood Clustering:** for this clustering we relied on K-Means due to our size of dataset. We choose features such as crime rate, education, depression, and alcohol consumption to compare them for different areas. We defined it to divide it into 3 clusters according to the elbow method. To identify safety patterns across neighborhoods, we applied K-Means clustering with k=3. The model grouped areas based on features crime rate (Crimes_m), poor health, alcohol consumption (Alcohol_C), depression levels, poverty rate (100_FPL_Pct), and education level (Edu_LessHS_Pct). The result was a safety zoning map categorizing regions into three risk levels: Low Risk (Safe), Moderate Risk, and High Risk.



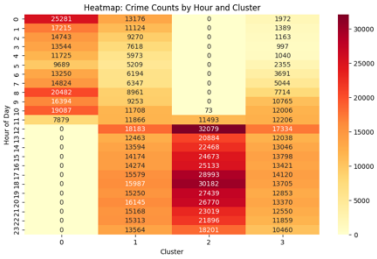
   b) **Time-based Clustering:** In time-based clustering we used k_means with 4 clusters. The main features that we used were TIME OCC, DATE OCC, Crm Cd so we can identify most common time for crimes, the crime code and whether it occurred on weekday or weekend.

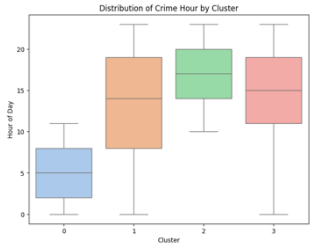
Field of computer Science & Engineering

Data Mining

GALALA UNIVERSITY
Powered by Arizona State University®

The analysis used K-Means clustering to group crimes based on time-related factors like hour, date, and type, identifying four key time patterns. The results showed that Cluster 1 had the highest crime volume with around 250,000 incidents, making it the most critical time window. Clusters 2 and 3 revealed specific peak times where Cluster 2 crimes concentrated around midnight, while Cluster 3 crimes peaked in late evening or early morning. These insights help law enforcement prioritize patrols and resources during high-risk periods to reduce crime effectively.





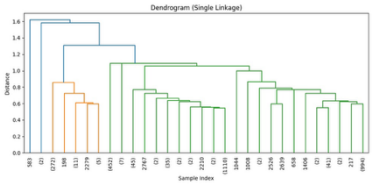

```
from sklearn.preprocessing import StandardScaler
time_features = data_cleaned[['Hour_OCC', 'Weekday_OCC', 'Crm Cd']].dropna()

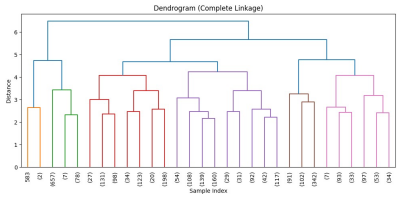
scaler = StandardScaler()
X_time = scaler.fit_transform(time_features)
```

```
from sklearn.cluster import KMeans

kmeans_time = KMeans(n_clusters=4, random_state=42)
data_cleaned['time_kmeans_cluster'] = kmeans_time.fit_predict(X_time)
```

c) **Victim Profiling Clustering:** This analysis used hierarchical clustering to uncover patterns among crime victims based on demographics and crime types. A sample of 3,000 cases was analyzed using four key features: victim age, ethnic descent, gender, and crime description. Text-based features were label-encoded, and all variables were standardized to ensure balanced input. Using both single and complete linkage methods, we constructed dendrograms to visualize natural groupings and identified three distinct victim clusters.






```
from scipy.cluster.hierarchy import linkage, dendrogram, fcluster
import matplotlib.pyplot as plt

Z = linkage(X_scaled, method='single')
plt.figure(figsize=(10, 5))
dendrogram(Z, truncate_mode='lastp', p=30, leaf_rotation=90., leaf_font_size=10.)
plt.title('Dendrogram ')
plt.xlabel('Sample Index')
plt.ylabel('Distance')
plt.tight_layout()
plt.show()
sample_data['Victim_Cluster'] = fcluster(Z, t=3, criterion='maxclust')
```

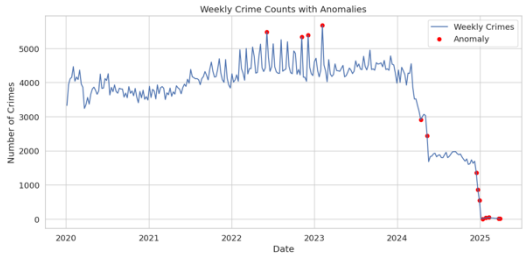
4) **Time-Series Analysis:**

a) **Anomaly Detection:** The aim was to determine what are the unusual spiked that occurred over time by dividing the crime count into the weeks they occurred in. For this, isolation forest was used then the results were further analyzed according to top crimes, their hourly trend and neighborhood hotspots.

Field of computer Science & Engineering

Data Mining

GALALA UNIVERSITY
Powered by Arizona State University®



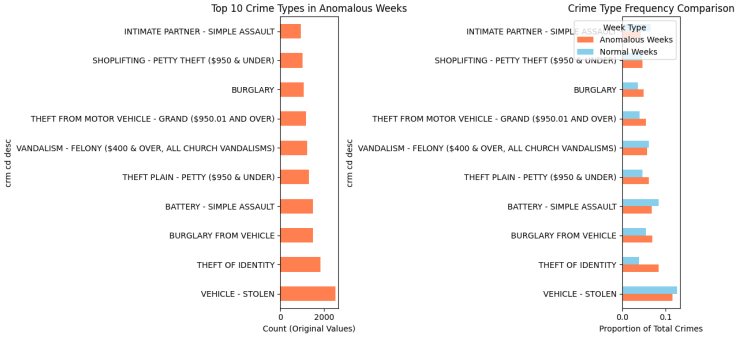

```python
from sklearn.ensemble import IsolationForest
import pandas as pd
from matplotlib import pyplot as plt
import seaborn as sns

data_pred_t = data_pred.copy()

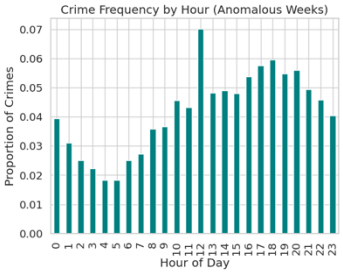
week_crime = data_pred_t.groupby(pd.Grouper(key='date occ', freq='W')).size().reset_index(name='crime_count')

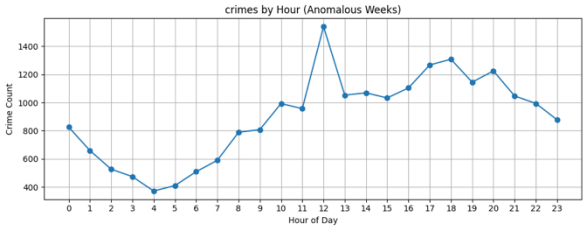
model = IsolationForest(contamination=0.05)
week_crime['anomaly_score'] = model.fit_predict(week_crime[['crime_count']])
week_crime['anomaly'] = week_crime['anomaly_score'].map({1: 0, -1: 1})
```

i) **Top Crime type:** As shown in the figure below, it was determined in analogous weeks the crime that was the most common was stolen vehicle which was also the same in normal weeks. However, the second most common was theft of identity while in normal weeks it was simple assault**.**



ii) **Hourly Trend:** Afterwards, the crimes that occurred on anomalous week were analyzed based on the hour that they occurred in. For instance, as shown below most crimes occurred at 12 pm and were most often in the afternoon.



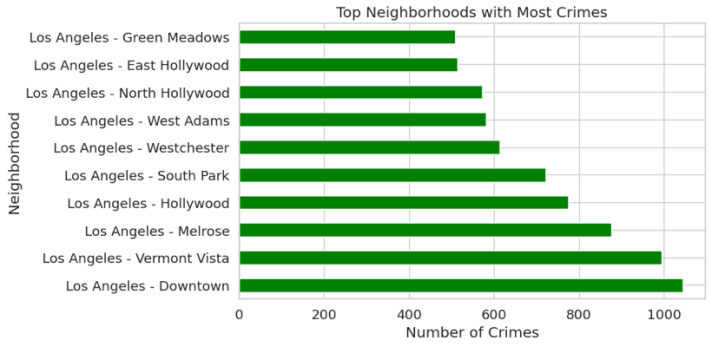


iii) **Neighborhood Hotspots:** Afterwards, the most common neighborhoods were crime happen in anomalous week was determined to be Downtown.

Field of computer Science & Engineering

Data Mining

GALALA UNIVERSITY
Powered by Arizona State University®

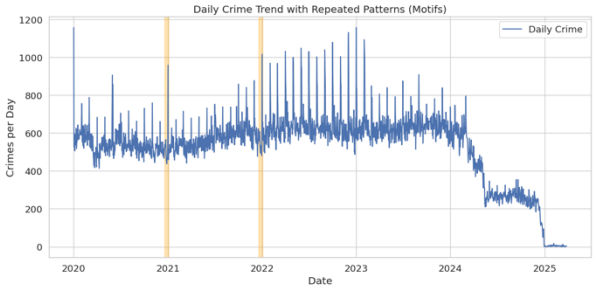


b) Motif Discovery: Afterwards, we wanted to detect the patterns that show up together when it comes to crime occurrences. First, the crime counts where grouped based on the date of occurrences. Then, we detect the most patterns that show weekly similarities using the stumpy library.

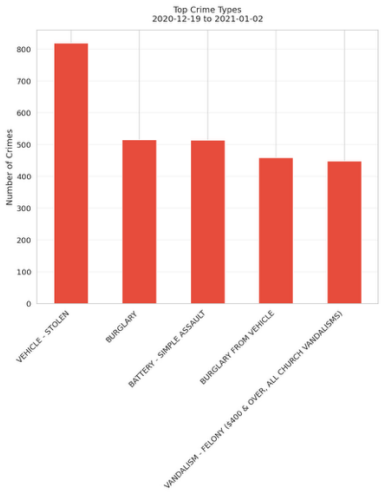
```
import stumpy
import numpy as np

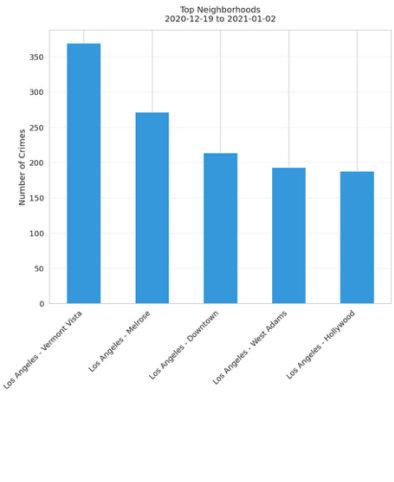

daily_crimes = data_pred_t.groupby(pd.Grouper(key='date occ', freq='D')).size().reset_index(name='crime_count')
daily_crimes = daily_crimes.set_index('date occ').asfreq('D').fillna(0).reset_index()
window_size = 14
mp = stumpy.stump(daily_crimes['crime_count'].astype(np.float64).values, m=window_size)


motif_idx = np.argsort(mp[:, 0])[:2]  # top 2 motifs
motif_dates = [daily_crimes['date occ'][i] for i in motif_idx]
```

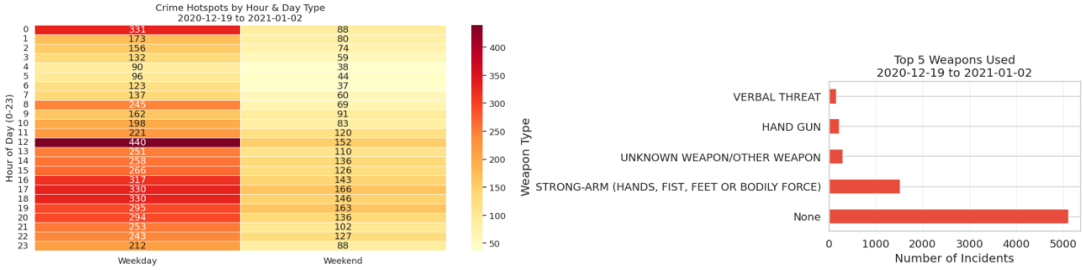


By using the top motif, we determined the most common crime type that occurred in weekly pattern was theft of identify and stolen vehicle. They mostly occurred in Vermount Vista and then in Melrose.

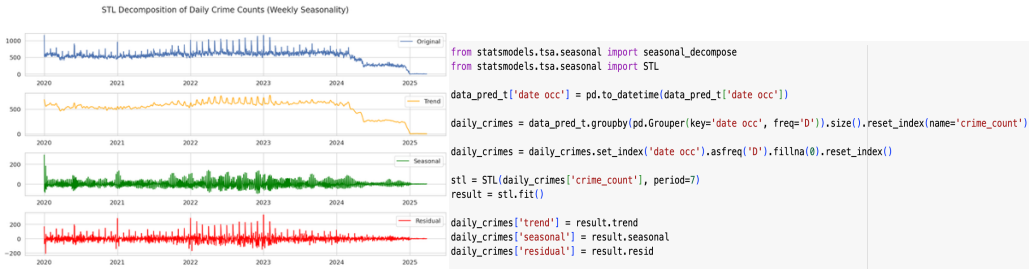




It was determined that in the most common patterns, crime occurred on weekdays on 12pm. Additionally, most crimes used then did not include a weapon.



c) **Seasonality Detection:** STL was applied to determine the trend which shows how crimes changed across the years, the seasonal which shows the most recurring patterns in the week, and the residual shows the anomalies or the noise that don't follow the trend. From that, we can determine that crime increased as years progressed with sudden peaks between 2022 and 2023 and that there are some repeating patterns within those years. Additionally, most repeating patterns here occurred on Fridays and Saturdays on days such as new year's and Memorial Day and in summer which are considered holidays in the US. The valleys showed that it had less crime on days after new years and midweeks.




```
from statsmodels.tsa.seasonal import seasonal_decompose
from statsmodels.tsa.seasonal import STL

data_pred_t['date occ'] = pd.to_datetime(data_pred_t['date occ'])

daily_crimes = data_pred_t.groupby(pd.Grouper(key='date occ', freq='D')).size().reset_index(name='crime_count')

daily_crimes = daily_crimes.set_index('date occ').asfreq('D').fillna(0).reset_index()

stl = STL(daily_crimes['crime_count'], period=7)
result = stl.fit()

daily_crimes['trend'] = result.trend
daily_crimes['seasonal'] = result.seasonal
daily_crimes['residual'] = result.resid
```

5) **OLS Regression:** to further understand how the socioeconomic features and demographic features where connected to the crime rate OLS regression was applied by making the crime rate the dependent variable.

```
economic_cols = ['mhi', '100_fpl_pct', '200_fpl_pct', 'sevhouse_b', 'overcrowding', 'homeless']
crime_cols = ['crimes_m', 'overall_mr']

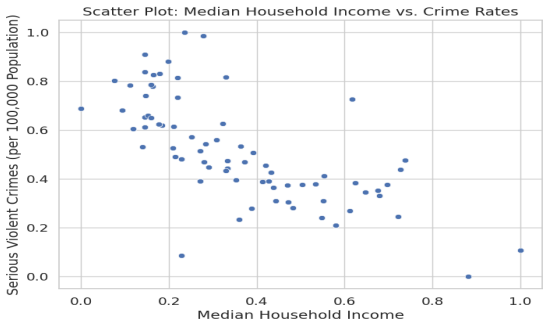
economic_crime_data = data_pred_t[economic_cols + crime_cols]

X = data_pred[economic_cols]
y = data_pred['crimes_m']

X = sm.add_constant(X)
model = sm.OLS(y, X).fit()
print(model.summary())
```

The socioeconomic features that were tested, where features like homelessness, overcrowdings, poverty, and income. The results show that the r-squared value was 0.64 which shows that the

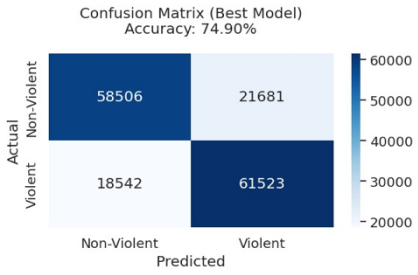
features do affect the crime rate. Additionally, it showed that as poverty, house burden, and homelessness increase that the crime rate also increases while the other features such as overcrowdings decreases.



The demographic features included that ethnicities of the population in each area. The R-value was determined to be 0.685 which shows that there is a strong relation between them. The results show that there is a strong correlation in order between the population of Latinos, white, black and then Asian with the crime rate.

6) **Predictive Modeling:**

a) **Crime Severity:** First, sampling techniques were used to ensure fare modeling prediction and then the LGBMC classifier was used with random search for hyperparameter tuning. The final results showed an accuracy of 75 percent by using the following parameters min_child_samples = 20, max_depth=6, and learning_rate =0.1.

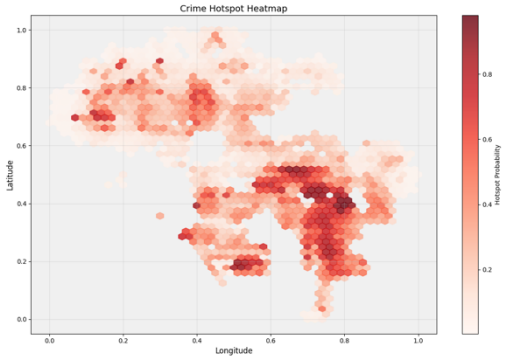


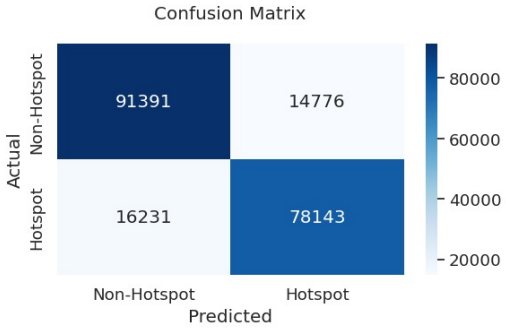

```
params = {
    'max_depth': np.arange(2, 7),
    'learning_rate': [0.01,0.05, 0.1, 0.2],
    'min_child_samples': [20, 50, 100]
}

model = LGBMClassifier(
    n_estimators=70,
    class_weight='balanced',
    random_state=42
)

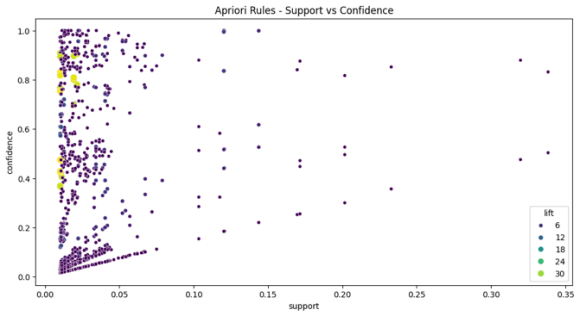
search = RandomizedSearchCV(
    model, params, n_iter=20,
    cv=3, scoring='f1', n_jobs=-1, verbose=1
)
search.fit(x_train, y_train)
print(search.best_params_)
```

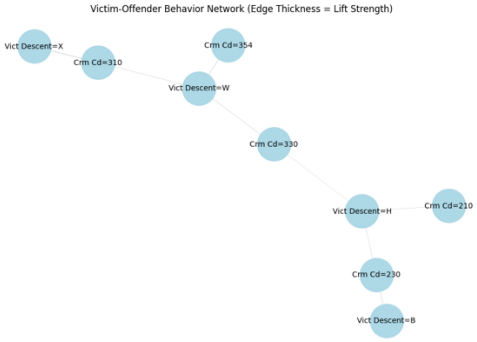
b) **Hotspot Prediction:** to predict whether an area is a hotspot or not, we first had to add another binary feature which will be our target and determine whether an area is a hotspot or not. The was done by dividing the areas into 500m grids and labeling it as hotspot if it is in the top 10% of crime count. Two other features were engineered in which "avg_dist_5_nearest" was to determine how dense the crimes are clustered in an area by finding the distance of other 5 crimes that are close to it. The second feature is "crime_count_7d" determines how many crimes occur in the same area in the last 7 days. The model used is HistGradientBoostingClassifier and it reached an accuracy of 82%. SHAP analysis was preformed where it was determined that avg_dist_5_nearest greatly affected the prediction and reaching an accuracy of 85%.



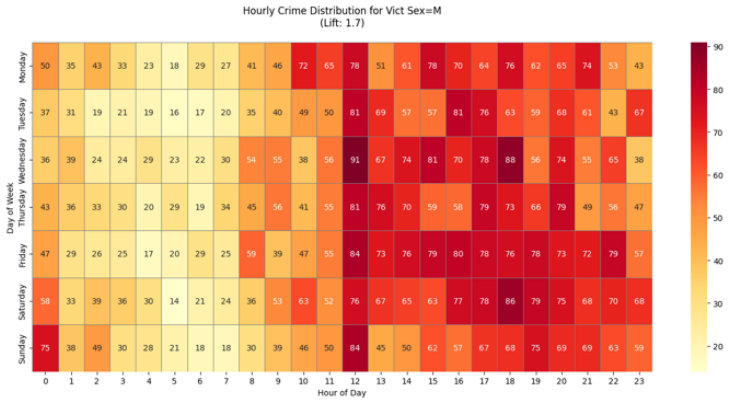


7) **Association Rule Mining:** the aim now is to find the common and frequent patterns that occur with a crime. That was done by first selecting the features that will be searched on such as the victim age, victim sex, and the weapon used. Afterwards, we turned the crime occurrence to transaction so the apriori algorithm can deal with them. The results show that female victims commonly show up with verbal threats also the adults are closely associated with verbal threats. Additionally, it was determined that crimes with code such as 510 frequently does not have the victim descent written down.





From there we were able to determine elements such as what time and days does crimes usualy occur again Men as shown below. It was concluded that it usually occurs in the afternoon and evening on Saturday and Wednesdays the most.



Conclusion:

We successfully achieved our objectives by finding what patterns occur the most when it comes to crimes in LA. Additionally, we were able to find the crime rate connection to socioeconomic and demographic features by utilizing clustering, descriptive analysis, time series analysis, and OLS regression. Furthermore, we were able to predict the severity of the crime and then predict which code does it follow. Then, we predicted whether an area is a hotspot or not reaching an accuracy of 85%. To conclude, while we were able to achieve our objectives, we aim to continue working on this issue and search for connections between rare crimes and find its patterns.